



**MAHALAKSHMI TECH CAMPUS
INSTITUTE OF ENGINEERS**

(A GROUP OF MAHALAKSHMI INSTITUTIONS)
Approved by AICTE & Affiliated to Anna University, Chennai.



PROJECT REPORT

SERVICE-NOW ADMINISTRATOR

ACADEMIC YEAR 2026-2027 (ODD SEMESTER)

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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**This Project Report is submitted for.....Degree/Semester
Practical Examination of Anna University conducted on**

INTERNAL EXAMINER

EXTERNAL EXAMINER

PROJECT ABSTRACT

The IT Procurement Automation project presents a structured workflow-based solution for managing the procurement of standard laptops in an organization. In a typical bulk procurement situation, the Human Resources department may raise a requirement for more than one hundred laptops for new employees, replacement needs, or planned expansion. The traditional process involves preparing requirement documents, contacting multiple vendors, collecting quotations, comparing commercial terms, negotiating with suppliers, obtaining authorization, preparing a purchase order, and communicating the final status to the relevant teams. When these activities are handled manually, the process can become difficult to track and may involve repeated follow-up, inconsistent records, and delays in approval or order creation.

The proposed system uses the ServiceNow platform and Flow Designer to organize these activities into a controlled digital workflow. HR can submit a procurement request containing the required quantity, standard laptop specification, required date, department, and business justification. Procurement users can maintain vendor information and record quotations received from different suppliers. The system keeps quotation information such as unit price, total amount, delivery period, warranty, validity, and commercial remarks in structured records. This provides a common information base for procurement review and negotiation.

A key design principle is that automation supports the procurement team rather than replacing human commercial judgment. Procurement officers review quotations, communicate with vendors, negotiate terms, and record the final vendor decision. After the required authorization is obtained, the workflow supports purchase order creation and sends notifications to relevant stakeholders. Flow Designer can automate task assignment, approval routing, status changes, and notifications, thereby reducing repetitive administrative work while maintaining human control over vendor evaluation and selection.

The project therefore demonstrates how ServiceNow can be applied to a real organizational process rather than only to a simple form submission. The resulting workflow provides traceability from the original HR requirement to the final purchase order, improves visibility of pending activities, maintains related records through reference fields, and creates a scalable foundation for future procurement enhancements. The implementation also introduces validation, role-based access, structured data management, testing, and reporting considerations required for a professional IT procurement workflow.

Project Flow: HR Requirement → Procurement Request → Vendor Quotations → Negotiation → Authorized Approval → Purchase Order → Notifications

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INTRODUCTION – PROCUREMENT PROBLEM STATEMENT

- IT procurement is a business process in which an organization acquires hardware, software, infrastructure, or related services required for its operations. Standard laptops are a common IT procurement item because organizations frequently purchase them in bulk for new employees, replacement programs, project teams, and office expansion.
- Although the item being purchased may be standardized, the supporting procurement process still involves several departments and external suppliers.
- HR identifies the business requirement, procurement coordinates with vendors, authorized personnel approve the commercial decision, and the final order is communicated to the supplier and internal stakeholders.
- In a manual environment, the same information may be repeated in emails, spreadsheets, documents, and individual follow-up messages. A procurement officer may maintain one file for vendor details, another for quotations, and another document for approval evidence.
- As the number of vendors and quotations increases, it becomes harder to determine which quotation is current, whether a negotiation has been completed, or whether the purchase order has already been created. Manual tracking also makes it difficult for management to obtain a real-time view of pending requests and overdue actions.
- The project addresses this coordination problem by treating the procurement request as the central record and connecting related vendor, quotation, task, approval, and purchase order information.
- The objective is not to automate the human decision of selecting a supplier.
- Instead, the system automates repeatable administrative steps and provides structured records so that procurement users can make decisions with complete information. This separation between workflow automation and human commercial judgment is important for maintaining accountability.
- The proposed solution is therefore a ServiceNow-based procurement workflow specifically designed for a bulk standard laptop scenario. It converts a sequence of loosely connected manual activities into a traceable process with defined states, responsibilities, validation rules, approvals, and notifications.
- The same design can later be adapted for other standard IT assets without changing the overall workflow concept.

EXISTING PROCESS AND ITS LIMITATIONS

- The existing procurement process normally begins when HR communicates a requirement to the procurement team. The requirement may include the number of laptops, standard configuration, department, required date, and a reason for the purchase.
- Procurement then identifies suitable suppliers and requests quotations. Vendors return quotations with commercial information such as unit price, taxes, delivery period, warranty, validity, and other terms. Procurement reviews these quotations and may negotiate with one or more vendors before reaching a final commercial decision.
- After the procurement team records its decision, the request is submitted to an authorized person for approval. Once approval is received, procurement prepares the purchase order and sends the order details to the selected supplier.
- HR and other relevant internal teams may also need to be informed. Each step creates information that should remain connected to the original requirement. In a manual process, however, the connection is often maintained through email subject lines, filenames, spreadsheet references, or human memory.
- The main limitations are lack of centralized visibility, repeated data entry, delayed follow-up, inconsistent status updates, difficulty locating approval evidence, and limited auditability.
- If a vendor sends a revised quotation, the procurement officer must ensure that the latest version is used in the decision.
- If an approval is delayed, another user may need to manually follow up. If a manager asks for the current status of ten requests, procurement must collect the information from several sources before preparing a response.
- The proposed system addresses these limitations by storing the lifecycle in related ServiceNow records and using Flow Designer to execute repeatable actions.

- The result is a controlled process in which the status of a request reflects the actual workflow stage and the responsible users can see the tasks that require attention.

Existing Manual Process vs Proposed Automated Process

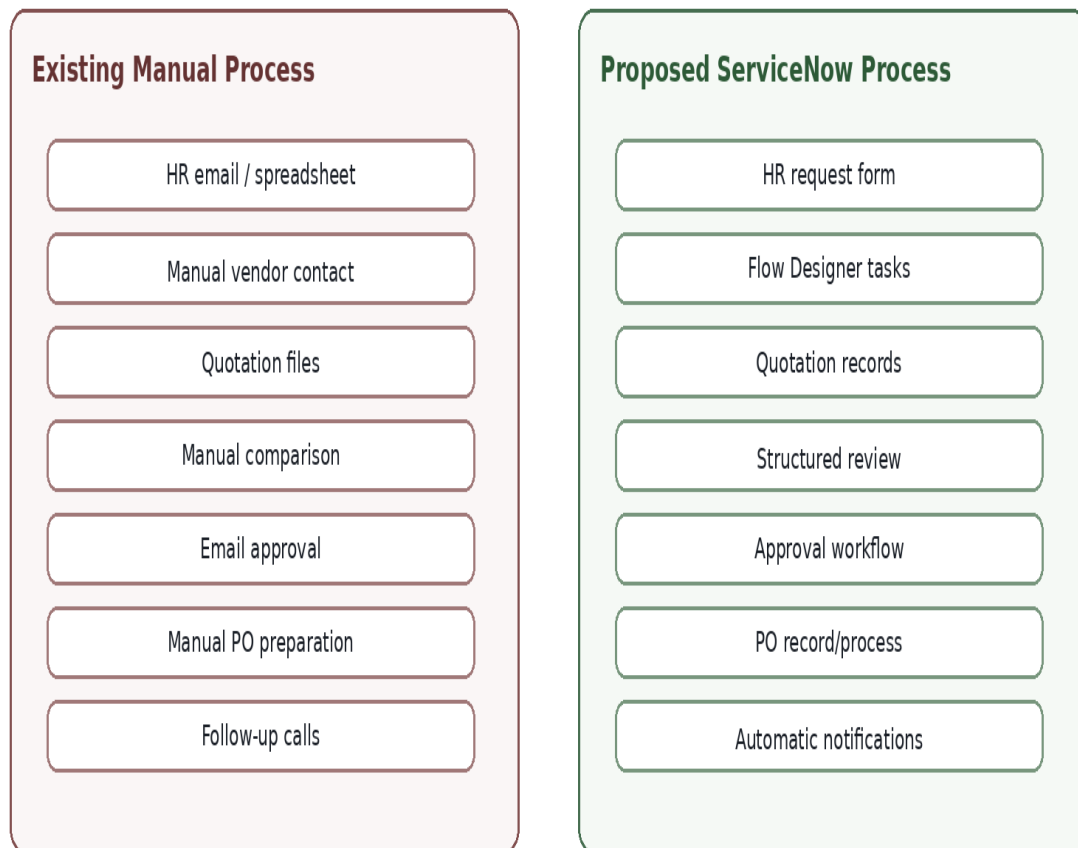


Figure 1.1 Existing manual process and proposed workflow-based process

PROPOSED SYSTEM

- The proposed system is a centralized IT procurement workflow implemented on ServiceNow. The process begins with the creation of a procurement request by HR or an authorized requester.
- The request captures the business need and technical requirement in a structured form.
- Once the request is submitted, Flow Designer can create or assign procurement tasks, update the request status, and notify the responsible group. Procurement users then maintain the supplier and quotation information associated with the request.
- The quotation stage is designed around structured comparison and human review. Multiple vendors can be associated with the same procurement request, and each vendor can have a quotation record containing the commercial terms received from that supplier.
- Procurement officers can update negotiation status and record agreed terms after discussions with vendors. The system does not automatically choose a supplier; it records the human decision and carries that decision forward into the approval and purchase order stages.
- When the procurement decision is ready for authorization, the workflow routes the request to the configured approver. Approval status and comments are stored as related records. A positive approval allows the process to continue to purchase order creation.
- If approval is rejected or returned for clarification, the request can be routed back to the responsible procurement user. After the purchase order is prepared, notifications can be issued to the supplier and internal stakeholders, and the request can be moved to its final tracking state.
- This architecture provides a clear boundary between data capture, workflow automation, human decision-making, and system administration.
- It also creates a reusable pattern for future procurement categories such as monitors, desktops, networking equipment, software licenses, and other standard IT assets.

End-to-End Procurement Lifecycle



PURPOSE AND SCOPE OF THE SYSTEM

- The purpose of the system is to streamline the procurement of standard laptops by replacing disconnected manual coordination with a structured workflow.
- The system provides a single record of the business requirement and connects it to the activities required to obtain quotations, review commercial information, complete negotiation, obtain authorization, create a purchase order, and communicate the result
- This improves process visibility without removing the procurement team's responsibility for commercial judgment.
- The scope includes bulk laptop requirement management, procurement request creation, vendor master management, quotation request and quotation recording, negotiation tracking, approval routing, purchase order information, notifications, task assignment, data validation, access control, and lifecycle tracking.
- The project also covers the ServiceNow tables required to represent these records and the Flow Designer actions required to connect them. The scope is intentionally focused on standard laptop procurement so that the workflow can be implemented and tested clearly.
- The project does not attempt to implement autonomous vendor selection, automatic commercial negotiation, physical inventory management at the warehouse, or direct payment settlement. These activities may be integrated in a future version.
- The current system concentrates on the procurement coordination layer and the information required to move a request from a business need to an approved purchase order.
- This controlled scope keeps the implementation realistic for an academic ServiceNow project while still demonstrating an end-to-end business workflow.
- The same framework can be extended after successful implementation. New procurement categories can reuse the request, approval, notification, and purchase order patterns while adding category-specific fields.
- Therefore, the laptop use case serves as a concrete implementation of a broader IT procurement automation model.

TARGET AUDIENCE AND BENEFITS

- The system is designed for HR employees, procurement officers, authorized approvers, vendors or vendor-facing procurement users, IT teams, management, and ServiceNow administrators.
- HR users initiate the business requirement. Procurement users manage suppliers, quotations, negotiations, approvals, and purchase orders. Approvers review and authorize the request.
- IT users may provide or validate technical specifications. Management can use status information for oversight, while administrators maintain the platform configuration and security model.
- The main benefit for HR is a clear submission and tracking mechanism. Procurement users benefit from centralized quotation and task information and reduced repetitive follow-up. Approvers receive structured information and a defined approval path.
- Management gains visibility into procurement status and outstanding activities. The organization benefits from consistent records, improved traceability, and a workflow that can scale to multiple requests.
- Vendors benefit indirectly because communication and order information are handled through a defined procurement process.
- The system also provides operational benefits. Mandatory fields reduce incomplete requests. Reference fields reduce duplicate supplier information. Automated notifications reduce the need for manual status emails. Workflow states provide a common vocabulary for progress.
- Approval records preserve evidence of authorization, and purchase order records retain the commercial result of the process. These features make the procurement lifecycle easier to monitor and easier to audit.
- Because the solution is built on ServiceNow, the organization can later extend the workflow with reporting, dashboards, integrations, document management, supplier portals, or additional procurement categories. The project therefore provides both immediate process improvement and a technical foundation for future expansion.

SYSTEM REQUIREMENTS – FUNCTIONAL REQUIREMENTS

- Functional requirements define what the procurement automation system must do. The central functional requirement is the ability to create and process a bulk laptop procurement request from initial HR requirement through purchase order creation.
- Each function is implemented as a combination of ServiceNow data structures, user interfaces, workflow actions, validations, and security controls.
- The procurement request function must capture the requester, department, laptop type, quantity, required date, specification, business justification, and current status.
- The vendor function must maintain supplier details and allow vendors to be associated with a request. The quotation function must store price and commercial terms for each supplier.
- The negotiation function must record the progress of commercial discussions without automatically deciding the supplier.
- The approval function must route the selected procurement decision to an authorized approver and store the approval result and comments.
- The purchase order function must store the approved supplier, quantity, agreed price, total amount, order date, and PO status.
- Notification functions must inform responsible users when a request is assigned, an approval is required, a status changes, or a purchase order is created.
- Tracking functions must allow authorized users to view the current state and related records.
- Administrative functions include role management, table and field configuration, workflow configuration, notification configuration, and access control.
- The system should also support basic reporting so that procurement users can identify open requests, pending approvals, quotation activity, and purchase orders.
- Together these requirements define a complete business workflow rather than an isolated form.

HR AND REQUESTER REQUIREMENTS

- The HR or requester interface is the starting point of the procurement lifecycle. The request form should be simple enough for a non-technical user to complete while still collecting the information procurement needs.
- The requester identifies the department, number of laptops required, standard laptop category, required date, and business justification. Technical specifications may be selected from predefined values or entered according to the organization's standard configuration.
- The form should prevent submission when critical information is missing. Quantity should be a positive number, required date should be valid, and the business justification should explain why the laptops are required.
- Where possible, reference fields should be used for department and requester rather than free-text entry. This reduces spelling variations and makes reporting more reliable. The request status should be controlled by the workflow instead of being freely edited by the requester.
- After submission, the requester should be able to view the request number, current status, and relevant progress information. The requester does not need access to every commercial field maintained by procurement.
- Role-based access can limit sensitive quotation or negotiation information while still allowing HR to see whether the request is pending, under procurement review, awaiting approval, or converted into a purchase order.
- The request record therefore acts as the anchor for the entire process. Every quotation, task, approval, and purchase order should contain a reference to this request.
- This design prevents the same business requirement from being recreated in multiple unrelated records and makes it possible to trace the complete lifecycle from a single starting point.

Illustrative Procurement Request Form Layout

Procurement Request	
Request Number PR-00125	Requester HR User
Department Human Resources	Laptop Type Standard Business Laptop
Quantity 100	Required Date 30-10-2026
Specifications Core i5 / 16 GB / 512 GB SSD	Business Justification New employee onboarding

PROCUREMENT TEAM REQUIREMENTS

- Procurement users require broader access because they are responsible for supplier coordination and commercial processing. They need to review submitted requirements, confirm that the requested specifications are complete, assign procurement tasks, identify appropriate suppliers, and record quotation information.
- The procurement interface should make it easy to see all open requests and the next action required for each request.
- For each request, procurement users should be able to associate multiple vendors and quotations. A quotation should capture the vendor, unit price, total amount, delivery period, warranty, validity period, negotiation status, and commercial remarks.
- When a vendor provides a revised quotation, the procurement user should be able to record the updated terms in a controlled manner so that the current commercial position is visible. The system should preserve enough history to support later review.
- The procurement user is also responsible for negotiation and final vendor selection. The system can provide structured fields for negotiation status, agreed price, comments, and decision date, but it should not automatically rank or select vendors.
- This ensures that commercial considerations, supplier relationships, technical suitability, delivery commitments, and organizational policy remain under human responsibility.
- Once the procurement decision is complete, the user initiates or triggers the approval stage. After authorization, the procurement user confirms the purchase order information and completes the final communication steps.
- The system should minimize repetitive entry by reusing information already stored in the request and selected quotation records.

ADMINISTRATIVE REQUIREMENTS AND USER ROLES

- System administration is required to maintain the reliability of the workflow. The ServiceNow administrator configures tables, fields, forms, roles, ACLs, UI Policies, Client Scripts, Flow Designer flows, notifications, and other system properties.
- Administration should follow controlled change practices so that modifications do not unintentionally affect active procurement records.
- The role model separates responsibilities. HR or requester users can create and track requests. Procurement users can manage vendors, quotations, negotiations, and purchase orders.
- Approvers can review and act on approval records. IT users can provide technical input where required. Administrators have configuration access.
- These responsibilities can be implemented through ServiceNow roles and ACLs so that users see and modify only the information required for their work.
- The role design also supports accountability. A procurement record can show who created or updated it, while an approval record can show the authorized person and decision date.
- This makes the workflow more transparent and discourages uncontrolled editing of important commercial information. The system administrator can review access requirements and adjust roles when organizational responsibilities change.
- The project therefore treats security as part of the functional design rather than as a separate afterthought. The user interface, data model, workflow, and access controls should all support the same responsibility model.

User Roles and Responsibilities

HR	Raises bulk laptop requirement	Request creation, tracking
Procurement	Handles vendors and quotations	Review, negotiation, selection
Approver	Authorizes purchase	Approve / reject
Admin	Maintains platform	Roles, tables, flows, ACLs
Vendor	Provides quotation	Price, delivery, warranty

NON-FUNCTIONAL REQUIREMENTS

- Non-functional requirements define the quality characteristics expected from the system. Performance is important because procurement users should be able to open forms, search records, and update workflow data without unnecessary delay.
- Workflow actions should execute reliably, and notifications should be generated promptly after significant events. The system should avoid unnecessary client-side processing and excessive record queries.
- Scalability is required because the organization may have multiple laptop procurement requests at the same time. The design should support many vendors, quotations, tasks, approvals, and purchase orders without requiring a different workflow for every request.
- Reusable Flow Designer actions, reference fields, controlled status values, and standardized forms support this objective.
- Security requires role-based access, ACLs, controlled data visibility, and appropriate protection of sensitive commercial information.
- Usability requires clear labels, logical form layouts, meaningful status values, and minimal unnecessary data entry. Reliability requires that workflow transitions and notifications behave consistently and that failed actions can be identified and corrected.
- Data integrity is particularly important because the purchase order must correspond to the approved commercial decision. Mandatory fields, reference fields, validation rules, and controlled state transitions reduce inconsistent records.
- Auditability requires that the system retain sufficient information about who performed important actions and when those actions occurred. These requirements define the quality standard for the implementation.

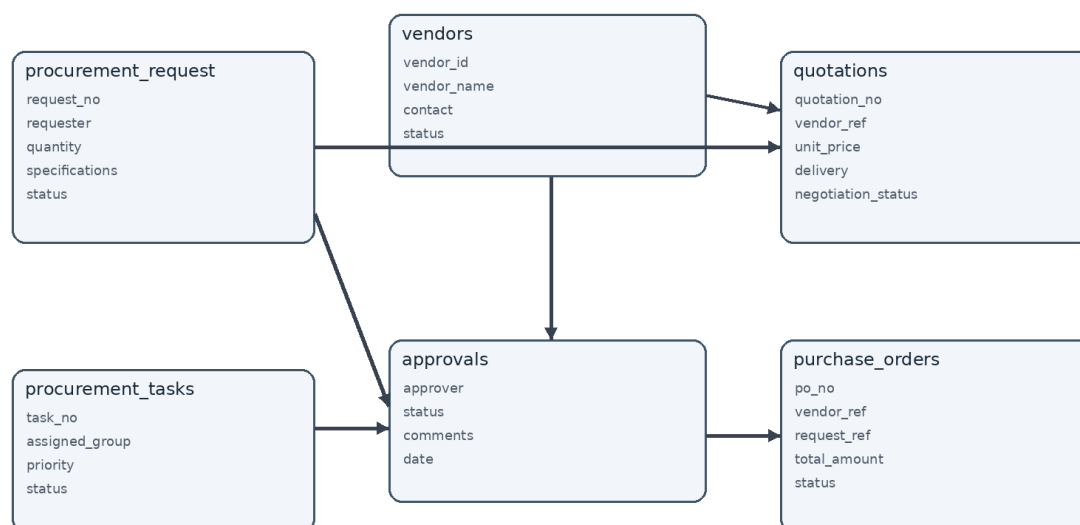
HARDWARE, SOFTWARE AND NETWORK REQUIREMENTS

- Because the application is delivered through the ServiceNow platform, the client-side hardware requirement is relatively modest.
- HR, procurement, approvers, and administrators require a standard computer or laptop capable of running a modern supported web browser.
- A keyboard and pointing device are sufficient for normal record entry. A larger display can improve productivity for procurement users who compare quotations or review multiple records, but it is not a mandatory system dependency.
- The software requirement includes a supported web browser and access to the organization's ServiceNow instance.
- The application itself uses ServiceNow Forms, Lists, Workspace features where applicable, Flow Designer, tables, roles, ACLs, Client Scripts, UI Policies, Business Rules, and Notifications.
- No separate Node.js, React, MongoDB, or custom server is required for the core implementation because the platform provides the application and database services.
- The network requirement is secure access to the ServiceNow instance over the organization's approved network path. HTTPS should be used for communication with the platform.
- Users should have stable connectivity because the workflow is browser-based and record updates are sent to the ServiceNow instance.
- If the organization uses VPN or identity federation, those controls should be integrated according to its infrastructure policy.
- The system should also be supported by administrative practices such as regular configuration backup or update procedures, controlled role assignment, testing in a non-production environment, and documented deployment steps.
- These operational requirements are important for maintaining a reliable workflow after the academic project is completed.

DATABASE SCHEMA AND DESIGN

- The database design represents the procurement lifecycle as a set of related ServiceNow tables. The central table is procurement_request because it represents the original business requirement. Vendor records are maintained separately so that the same supplier can participate in multiple procurement requests.
- Quotation records are linked to both the procurement request and vendor, allowing several suppliers to submit commercial information for the same requirement.
- Procurement tasks represent work assigned to procurement users or groups. Approval records represent authorization decisions and contain the approver, status, comments, and decision date.
- Purchase order records contain the final commercial result and reference the procurement request and selected vendor. Reference fields connect these records so that users can navigate from the request to its related activities without duplicating the complete data in every table.
- The design also uses controlled fields to maintain data integrity. Mandatory fields ensure that important information is not omitted. Reference fields ensure that related values come from valid records.
- Choice fields can restrict status values to approved states such as Draft, Submitted, Procurement Review, Quotation, Negotiation, Approval, PO Created, and Closed. Validation rules can prevent invalid quantities, dates, or financial values.
- The database is therefore not merely a storage layer. It is the structural foundation that allows the workflow to maintain relationships, support reporting, and preserve traceability throughout the procurement lifecycle.

Logical Data Model - Procurement Records



DATABASE TABLES AND DATA DICTIONARY

- The following tables form the core data model for the standard laptop procurement scenario. The exact field names can be implemented as custom ServiceNow table fields according to the organization's naming standards.
- The important design principle is that each table has a clear responsibility and uses references rather than repeated free-text information wherever possible.
- The procurement_request table stores the business requirement and workflow status. The vendors table stores reusable supplier master information.
- The quotations table stores supplier-specific commercial terms. The procurement_tasks table records operational work. The approvals table preserves authorization evidence.
- The purchase_orders table records the final approved order. Together these entities provide a complete record chain.
- Financial and date fields should use appropriate ServiceNow data types and validation. Quantity should be numeric, monetary values should use currency-compatible fields where appropriate, and required dates should use date or date-time fields.
- Reference fields should point to the relevant request, user, vendor, or group records.
- Status fields should be controlled so that reporting remains consistent across requests.
- This structure supports future reporting because management can count requests by status, compare quotation activity, identify pending approvals, and summarize purchase orders without reconstructing the lifecycle from email messages.

Table	Important Fields	Purpose
procurement_request	Number, requester, department, quantity, specification, required date, status	Stores the original HR requirement
vendors	Name, contact, email, phone, address, status	Maintains supplier master information
quotations	Request, vendor, unit price, total, delivery, warranty, validity, negotiation status	Stores supplier commercial terms
procurement_tasks	Request, description, group, priority, status, due date	Tracks assigned procurement work
approvals	Request, approver, status, comments, date	Stores authorization evidence
purchase_orders	Request, vendor, quantity, agreed price, total, order date, status	Stores approved purchase order information

IMPLEMENTATION – SYSTEM ARCHITECTURE

- The implementation combines the ServiceNow user interface, application tables, workflow automation, notifications, and access control into one architecture. A user begins at the presentation layer through a ServiceNow form or list. The request is stored in the procurement_request table.
- Flow Designer observes the configured trigger and executes actions such as creating tasks, assigning groups, updating states, requesting approval, and sending notifications.
- Related records are stored in the vendor, quotation, task, approval, and purchase order tables.
- The architecture is intentionally modular. The user interface is responsible for data capture and presentation. Client Scripts and UI Policies provide immediate form behavior and validation.
- Server-side components such as Business Rules can enforce additional logic where appropriate. Flow Designer provides orchestration across records and groups.
- The database layer provides persistence and relationships. Security controls apply across the application so that the same workflow can be used by different roles without exposing unnecessary data.
- This separation makes the implementation easier to test. Form validation can be tested independently from workflow execution. Database relationships can be tested by creating related records. Flow Designer can be tested using representative procurement requests.
- Notifications can be verified after state changes. Role and ACL tests can confirm that restricted commercial information is not visible to unauthorized users.
- The architecture also supports future integration. An external vendor portal, document repository, enterprise identity provider, or purchasing system can be connected later without redesigning the core procurement request model.

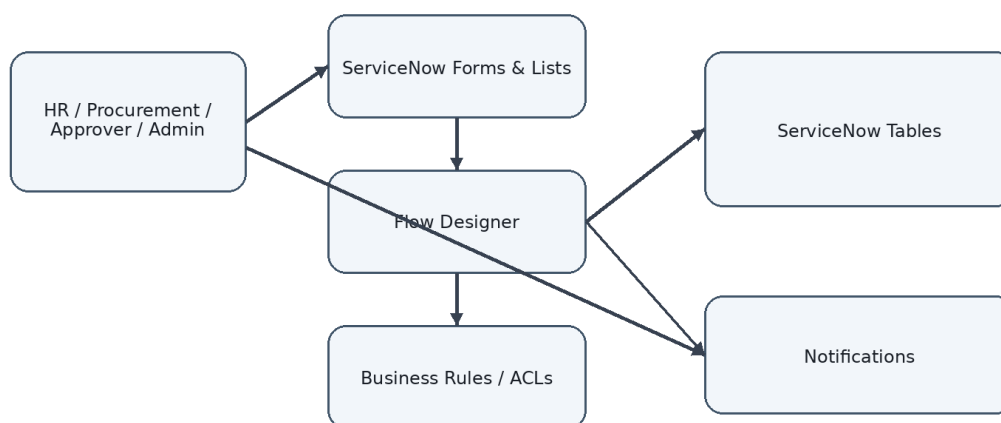
ServiceNow IT Procurement Automation - System Architecture

Figure 3.1 ServiceNow IT procurement automation architecture

3.1 PROCUREMENT REQUEST FORM IMPLEMENTATION

- The procurement request form is implemented as the first user-facing component. It should present fields in a logical order: requester and department, laptop category, quantity, specification, required date, business justification, and status.
- System-generated fields such as request number and created date should not require manual entry. The form layout should make it clear which information is mandatory and which information will be populated or controlled by the workflow.
- Client Scripts can validate values at the time of entry. **For example**, a quantity of zero or a negative quantity should not be accepted. A required date that is invalid for the organization's process can be rejected. UI Policies can make fields mandatory, visible, or read-only according to the current state.
- After submission, status fields can become controlled by the workflow so that users cannot bypass procurement stages manually.
- The form should also provide related lists or links to associated vendors, quotations, tasks, approvals, and purchase orders. This gives authorized procurement users a complete view of the request without navigating through unrelated modules.
- HR users can receive a simplified view that exposes progress without revealing sensitive commercial information.
- A well-designed form reduces data correction later in the process. The request should contain enough information for procurement to begin supplier communication without repeatedly contacting HR for basic details.
- At the same time, the form should avoid unnecessary fields that increase completion time without adding business value.

VENDOR MASTER AND RFQ PROCESS

- Vendor management begins after the procurement request has been reviewed and accepted for procurement processing. The vendors table stores reusable supplier information such as vendor name, contact person, email, phone number, address, and active status.
- Maintaining a vendor master prevents procurement users from typing different spellings of the same supplier into different requests and improves consistency in reporting.
- The procurement team identifies suitable suppliers according to the organization's sourcing policy and sends a request for quotation. The RFQ communication can be managed through configured notifications or approved external communication channels.
- The system records the fact that quotation information is expected from each supplier and links the supplier to the procurement request. This creates a structured starting point for quotation tracking.
- The RFQ process should make the requested specification clear and consistent across vendors. Each supplier should receive the same essential requirement information so that quotations are commercially comparable. Procurement can record the date of RFQ communication and the expected quotation response if such fields are included in the implementation.
- When a quotation arrives, the procurement user creates a quotation record linked to the relevant vendor and request.
- The vendor master and RFQ design separates supplier identity from individual commercial responses. A vendor may appear in many requests, while each quotation belongs to one procurement request and one vendor.
- This relationship is essential for maintaining a clean database and for later reporting on supplier participation.

VENDOR QUOTATION MANAGEMENT

- Quotation management is one of the most important parts of the procurement workflow because the commercial information received from vendors must be structured before it can be reviewed.
- The quotation record should contain the linked procurement request and vendor, quotation reference, unit price, total amount, delivery period, warranty, quotation validity, negotiation status, and commercial remarks.
- Additional fields can be added for tax, payment terms, configuration details, or quotation document attachment when required.
- When multiple vendors respond, each quotation is stored separately but remains linked to the same procurement request. Procurement users can open the related list and compare the records.
- The system should not automatically label one supplier as the winner. Instead, the procurement user reviews the quotations using the organization's commercial and technical criteria and records the human decision.
- This keeps the workflow auditable and avoids treating a simple price comparison as a complete procurement decision.
- If a vendor changes its price or delivery commitment during negotiation, the updated information should be recorded using controlled editing or a revised quotation record according to the organization's chosen design.
- The quotation status should indicate whether it is received, under review, under negotiation, accepted for approval, or not selected. This makes the state of each supplier response visible.
- Structured quotation data also enables later reporting. Procurement management can identify how many vendors responded, how many quotations are pending, how long quotation review takes, and how often negotiations change the commercial terms.
- These insights are difficult to obtain reliably from unstructured email correspondence.

NEGOTIATION AND HUMAN VENDOR SELECTION

- Negotiation is deliberately treated as a human-controlled stage. The procurement officer may discuss price, delivery, warranty, payment terms, configuration, or service commitments with vendors.
- These discussions can affect the commercial suitability of a quotation, and therefore the system should provide a place to record negotiation status and the final agreed terms rather than attempt to perform the negotiation automatically.
- A practical implementation can use fields such as negotiation status, last negotiation date, negotiated unit price, agreed delivery period, negotiation remarks, and final decision.
- When a quotation is under negotiation, the procurement task remains open until the procurement user completes the activity. The workflow can remind or notify the responsible user, but the commercial conversation itself remains outside the automated decision logic.
- Once the procurement officer completes the review, the final vendor selection is recorded as an explicit business decision.
- The selected vendor reference can then be used by the approval stage and purchase order stage. Other vendor quotations remain linked to the request so that the organization retains a record of the alternatives considered.
- This approach provides a useful balance between automation and governance. ServiceNow automates the administrative path around the decision, while procurement professionals retain responsibility for evaluating suppliers and negotiating commercial terms.
- The resulting audit trail shows the original quotations, negotiation information, selected supplier, approval, and final purchase order.

Approval Decision Flow

Figure 3.4 Human-controlled review, decision and approval path

APPROVAL PROCESS IMPLEMENTATION

- Approval is a control point between commercial decision-making and order placement. After the procurement officer records the final vendor decision and agreed terms, the request enters an approval state.
- Flow Designer can create an approval action for the configured authorized approver or approval group. The approval record stores the approver, current status, comments, and approval date, providing explicit evidence that the organization authorized the purchase.
- The approval interface should present the information required for a meaningful decision. This may include the original business justification, quantity, laptop specification, selected vendor, agreed unit price, total amount, delivery period, warranty, and relevant procurement comments.
- The approver can then approve, reject, or return the request for clarification depending on the configured process.
- If approval is rejected, the workflow should update the procurement request and notify the procurement user so that the issue can be addressed.
- If the approval is granted, the workflow can move the request to the purchase order stage. If the organization requires multiple approvals based on purchase value or department, Flow Designer conditions can route the request to the appropriate approval path.
- Approval automation reduces the risk of placing an order without authorization while also reducing manual follow-up.
- The system records the decision as a workflow event, allowing authorized users to see whether a request is waiting for approval, approved, rejected, or returned for clarification.

PURCHASE ORDER PROCESS

- Purchase order processing begins after the required authorization has been obtained. The purchase order record should reference the procurement request and selected vendor and contain the approved quantity, agreed unit price, total amount, order date, and PO status.
- Where required, additional fields can store delivery address, payment terms, tax details, or a reference to the approved quotation.
- The system should reuse information from the procurement request and selected quotation wherever possible. This reduces duplicate data entry and lowers the risk of creating a purchase order with terms that differ from the approved commercial decision.
- Validation should confirm that the request has an approved status and that a valid selected vendor and agreed amount exist before the PO is created or marked ready for issue.
- After the purchase order is created, Flow Designer can update the procurement request status and notify the procurement team, HR, and other relevant stakeholders.
- The vendor can receive the order through the organization's approved communication process. The PO status can progress through values such as Draft, Approved, Sent, Acknowledged, and Closed according to the actual business process.
- The purchase order stage completes the main automation path but does not have to represent physical delivery or asset inventory. Those activities can be integrated later.
- The current project focuses on the procurement transaction and ensures that the final order remains traceable to the original business requirement and approval.

FLOW DESIGNER ARCHITECTURE

- Flow Designer is the orchestration layer of the project. A flow is configured around a defined trigger and a sequence of actions.
- The trigger can be based on creation or update of a procurement request, a state change, or another business event. The actions then perform the repeatable work required for that stage, such as creating a task, assigning a group, requesting approval, updating a status, or sending a notification.
- The main procurement flow can begin when a new request is submitted. The first actions validate the request state and create a procurement task.
- The procurement team processes the request and records vendor quotations. When the procurement user indicates that quotation review is complete, the workflow can transition to negotiation or approval according to the configured conditions.
- After approval, the flow supports purchase order creation and final notifications.
- Flow Designer is particularly useful because the logic is visible as a sequence of steps rather than being hidden entirely inside custom code. Each action can be tested and reviewed by administrators.
- Reusable actions can be designed for common tasks such as notification, status update, or task creation. Conditions can branch the workflow when approval is rejected, when required data is missing, or when different approval levels are needed.
- The workflow should be designed with clear state transitions and exception paths. A flow that only handles the successful path is not sufficient for a real business process.
- Rejection, incomplete information, missing quotation responses, and failed notifications should be considered during implementation and testing.

Flow Designer - Example Procurement Automation

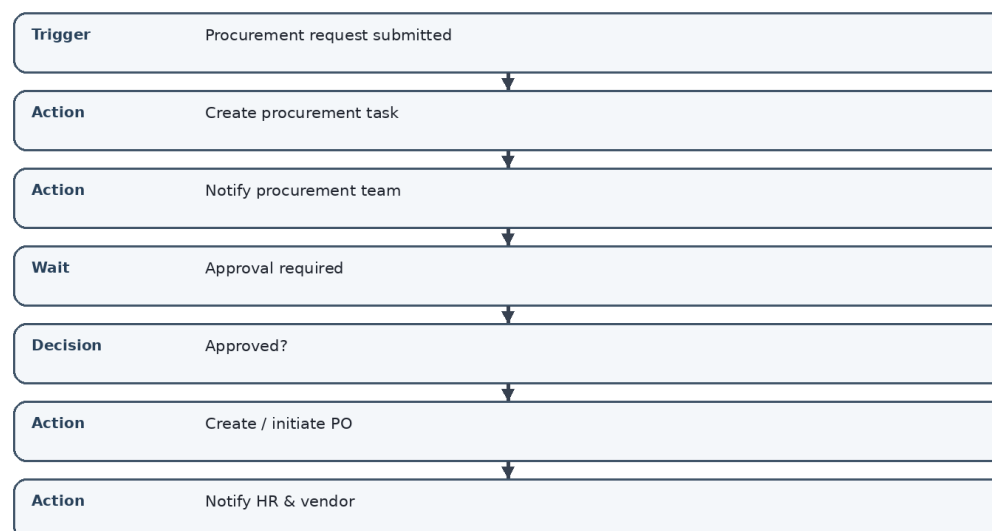


Figure 3.7 Flow Designer automation sequence

FLOW TRIGGERS AND ACTIONS

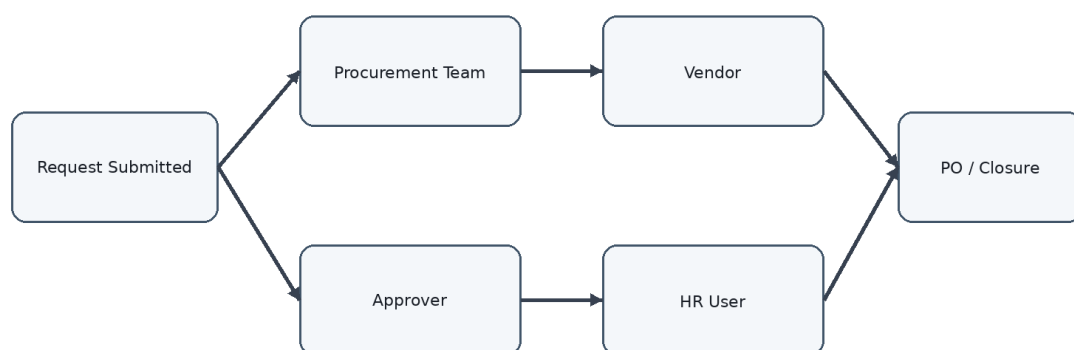
- The trigger defines when automation begins. A suitable primary trigger for the project is submission of a procurement request with the required fields completed.
- Additional flows or subflows can be triggered when the request reaches procurement review, when a procurement decision is recorded, when an approval is completed, or when a purchase order is created.
- Typical actions include Create Record, Update Record, Ask for Approval, Create Task, Send Email, and Wait for Condition.
- **For example**, after request submission the flow can create a procurement task and notify the procurement group. When approval is required, the flow can create the approval and wait until a decision is recorded. After approval, it can create or prepare the purchase order and send the relevant notifications.
- Flow inputs and outputs should be clearly defined. The procurement request record should be available to downstream actions, while the selected vendor and quotation should be available when the approval or PO stage is reached. Reusing data from previous steps prevents unnecessary record queries and reduces duplication.
- Where a reusable action is likely to be needed by several flows, it can be implemented as a subflow or action with documented inputs and outputs.
- Error handling is also part of the workflow design. If a required record cannot be found, the flow should not silently continue to create an incorrect purchase order.
- Instead, it should stop safely, record the problem, and notify an appropriate administrator or procurement user. This makes the automation more trustworthy and easier to support.

CONDITIONS AND BRANCHING

- Conditions control how the workflow responds to different business situations. For example, the system can check whether a procurement request contains a valid quantity and required date before allowing the process to continue.
- At the quotation stage, the workflow can check whether the procurement user has completed the required review information before moving to approval. At the approval stage, the workflow branches according to the approval decision.
- A simple approval branch can contain an Approved path and a Rejected path. The Approved path continues to purchase order processing, while the Rejected path updates the request status and notifies procurement.
- A third path can be used for Returned or Clarification Required if the organization needs the approver to send the request back without rejecting the business requirement completely.
- Conditional logic can also support value-based approval. For example, requests above an organizational threshold can be routed to an additional authority.
- The exact threshold should come from the organization's procurement policy rather than being invented by the application. The project demonstrates the technical mechanism for such branching while keeping policy values configurable.
- Good branching design avoids overly complex flows. Each condition should have a clear business reason, and the resulting states should be easy for users to understand.
- Status names should remain consistent across forms, reports, notifications, and test cases so that users do not have to interpret different terms for the same stage.

NOTIFICATIONS AND TASK ASSIGNMENT

- Notifications are used to keep participants informed when an action is required or when the status of a request changes. Typical events include new procurement task assignment, quotation review completion, approval request, approval decision, purchase order creation, and final status update.
- Notifications should contain the request number and enough contextual information for the recipient to identify the required action without exposing unnecessary sensitive data.
- Task assignment is equally important because automation should create clear ownership. When a request is submitted, a procurement task can be assigned to the procurement group. Individual task assignment can then be performed according to the organization's team structure.
- Tasks should have a description, priority, status, due date, and reference to the procurement request. This allows procurement managers to identify overdue or unassigned work.
- The system can use ServiceNow notifications to send email alerts based on record events or Flow Designer actions. Notification templates should be concise and consistent.
- Approval notifications should identify the request, selected vendor, amount, and required decision. Purchase order notifications should communicate the PO reference and current status to the intended recipients.
- Automated communication reduces repeated manual follow-up while preserving a clear audit trail. The notification itself is not the business record; the underlying task, approval, or procurement request remains the authoritative record. This distinction is important when designing the system and testing whether notifications correspond correctly to the underlying state.

Notification and Task Assignment Flow*Figure 3.10 Automated notifications and task assignment*

DATA VALIDATION AND CONTROL MECHANISMS

- Data validation is implemented at multiple levels. Client-side controls provide immediate feedback while the user is entering information. UI Policies can make fields mandatory or read-only depending on the state. Client Scripts can validate numeric or date values and can populate or clear fields where appropriate.
- These controls improve usability but should not be treated as the only security boundary.
- Server-side validation can be implemented using Business Rules or other platform controls where a rule must be enforced regardless of how the record is updated.
- **For example**, a purchase order should not be marked approved if the related request has not received the required authorization. A quotation should reference an active vendor, and a procurement request should have a valid requester and quantity.
- Controlled status values prevent users from creating inconsistent lifecycle states. Reference fields reduce duplicate names and improve relationships.
- ACLs restrict access to sensitive data. Audit history can help administrators identify changes to important records. Together these mechanisms provide defense in depth: the form guides users correctly, the workflow controls transitions, and server-side security prevents unauthorized changes.
- Validation should be tested for both positive and negative cases. A valid request should proceed normally, while missing or invalid information should produce a clear message and prevent the incorrect transition.
- Testing should also confirm that a user without the required role cannot bypass the intended controls.

END-TO-END IMPLEMENTATION PROCEDURE

- The implementation can be performed in a controlled sequence. First, create the procurement request and supporting tables with the required fields and reference relationships.
- Next, configure the forms and lists so that users can create and review records. Then create the vendor and quotation structures and confirm that multiple quotations can be linked to one procurement request. After the data model is stable, configure the approval and purchase order records.
- The next stage is workflow configuration. Create the Flow Designer flow with a clear trigger and actions. Add task creation and notification actions first, then add approval routing and purchase order processing.
- Configure conditions for successful, rejected, and clarification paths. Test each stage using a small set of representative records before testing the complete end-to-end process.
- Security configuration should then be applied. Define the user roles and ACLs, verify access to each table and field, and test the system with different user accounts.
- Finally, configure notifications, reports, and administrative views. The system should be tested in a development or test instance before any production deployment.
- A documented deployment procedure should identify the tables, fields, flows, scripts, UI policies, roles, ACLs, and notifications that form the application.
- This documentation is useful for academic evaluation as well as future maintenance. It also demonstrates that the project is designed as a maintainable application rather than a collection of isolated configuration steps.

TESTING – STRATEGY AND APPROACH

- Testing verifies that the configured ServiceNow components behave as intended and that the complete procurement lifecycle can be executed without inconsistent records. The project uses a layered strategy.
- Unit-level configuration checks validate individual forms, fields, scripts, UI policies, and tables. Workflow testing validates Flow Designer triggers, actions, conditions, approvals, and notifications.
- Integration-level testing validates relationships between request, vendor, quotation, approval, and purchase order records.
- Functional testing uses realistic procurement scenarios. A normal request should move through the complete process. Negative scenarios should verify that incomplete requests cannot progress.
- Approval rejection should stop purchase order processing. Multiple quotations should remain linked to the same request. Unauthorized users should not be able to modify restricted records.
- Notification testing should confirm that the correct recipient receives the correct message at the correct stage.
- User acceptance testing focuses on whether the workflow makes sense to the people who would operate it. HR users should be able to submit requirements without unnecessary complexity.
- Procurement users should be able to locate related vendor and quotation records quickly. Approvers should receive enough information to make an authorization decision. Administrators should be able to identify failed or unexpected workflow behavior.
- Testing results should be documented with test case ID, scenario, expected result, actual result, and status. Any failure should be corrected and the test repeated.
- This creates evidence that the project was validated systematically rather than only demonstrated through one successful scenario.

Testing Lifecycle for Procurement Automation



4.1 TEST CASES AND EXPECTED RESULTS

The following test cases cover the most important workflow paths. They are designed to verify both normal processing and exception handling.

The expected result for each test is tied to the business requirement so that the test demonstrates a meaningful system behavior rather than only a technical configuration check.

Test Case 1

Verifies that a valid HR request is created with a unique request number and Submitted state.

Test Case 2

Verifies mandatory-field validation.

Test Case 3

Verifies that a procurement task is created after submission.

Test Case 4

Verifies that multiple vendor quotations can be associated with one request.

Test Case 5

Verifies that negotiation information can be recorded without automatic vendor selection.

Test Case 6

Verifies that an approval is generated for the authorized approver.

Test Case 7

Verifies the rejected approval path.

Test Case 8

Verifies that an approved request can proceed to purchase order creation.

Test Case 9

Verifies that the PO contains the approved vendor, quantity, and agreed price.

Test Case 10

Verifies that notifications are generated for the configured events.

Test Case 11

Verifies role-based access.

Test Case 12

Verifies that related records remain connected throughout the lifecycle.

The test suite should be repeated after any major workflow or configuration change. This reduces regression risk and provides evidence that an update has not broken an existing procurement stage.

IT PROCUREMENT AUTOMATION – STANDARD LAPTOP ORDER

ID	Scenario	Expected Result
TC01	Submit valid laptop request	Request created with unique number and Submitted state
TC02	Submit without mandatory quantity	Submission blocked with validation message
TC03	Request enters procurement	Procurement task created and group notified
TC04	Record multiple vendor quotations	Each quotation links to the same request and correct vendor
TC05	Record negotiation outcome	Negotiation details saved; no automatic vendor selection
TC06	Send for approval	Approval record created for authorized approver
TC07	Reject approval	PO creation stopped and procurement notified
TC08	Approve request	Request moves to PO processing
TC09	Create PO	PO contains approved vendor, quantity and agreed amount
TC10	Trigger status notification	Correct notification generated
TC11	Access restricted record	Unauthorized update is prevented
TC12	Open lifecycle from request	Related quotation, approval and PO records are traceable

INTEGRATION TESTING AND USER ACCEPTANCE

- Integration testing checks that the individual components work together. A procurement request must be available to the Flow Designer trigger.
- The flow must create a task referencing the correct request.
- Vendor and quotation records must remain connected to that request. The approval action must use the selected vendor and agreed commercial information. After approval, the purchase order must reference the same request and vendor. Notifications must use the state and recipient information from the correct record.
- User acceptance testing should involve representative roles. An HR tester creates a requirement and confirms that the form is understandable.
- A procurement tester processes the request and enters several quotations. An approver reviews the request and records a decision. An administrator verifies the workflow history and access controls.
- The scenario is considered acceptable when each participant can complete the required responsibility without bypassing the intended process.
- The testing process should also verify data consistency after repeated updates. For example, if a quotation is changed during negotiation, the approval information should reflect the final agreed terms. If approval is rejected, a purchase order should not be created accidentally.
- If the request is returned for clarification, the workflow should not create duplicate tasks unless the process specifically requires them.
- Successful integration and user acceptance testing provide confidence that the application works as a business process rather than only as individual ServiceNow configurations.
- They also reveal usability issues that technical testing alone may not identify.

TECHNICAL CHALLENGES

- One technical challenge is translating a real procurement process into a workflow that is detailed enough to be useful but simple enough to maintain.
- Procurement contains many possible exceptions, such as incomplete requirements, delayed quotations, revised commercial terms, rejected approvals, and vendor changes. The project therefore uses defined states and controlled transitions rather than allowing every user to change every field at any time.
- Another challenge is maintaining relationships between records. The request, vendor, quotation, approval, task, and PO records must reference one another correctly.
- If a relationship is missing, users may see incomplete information and reporting may become unreliable. Reference fields and validation rules are therefore essential parts of the implementation.
- Workflow sequencing is also a technical challenge. Some actions must happen only after another action is complete.
- The purchase order should not proceed before approval, and approval should not proceed before the procurement decision is recorded.
- Flow Designer conditions and state checks provide the control mechanism, while testing confirms that the sequence works correctly.
- Security adds another layer of complexity. HR should not automatically receive the same commercial visibility as procurement. Approvers should be able to act on their assigned approvals without gaining unnecessary administrative privileges.
- Administrators need configuration access, but normal users should not. Roles and ACLs must therefore be designed together with the data model and user interface.
- Finally, workflow errors need to be diagnosable. If a notification fails or a flow stops because a related record is missing, administrators need enough information to identify the cause.
- Clear field names, meaningful statuses, task descriptions, and documented flow logic make troubleshooting easier.

DESIGN CHALLENGES AND USABILITY CONSIDERATIONS

- The first design challenge is balancing completeness with usability. A procurement form must collect enough information for the procurement team without overwhelming HR users with fields that they do not understand.
- The solution is to separate general business requirements from procurement-specific information. HR enters the requirement, while procurement users maintain supplier and commercial fields.
- The second challenge is representing the procurement lifecycle clearly. Users should be able to understand what Submitted, Procurement Review, Quotation, Negotiation, Approval, and PO Created mean.
- The workflow therefore uses a controlled state model and task assignments. Each state should correspond to a real business activity and should have a responsible user or group.
- The third challenge is presenting related information. Procurement users may need to review several quotations while working on one request. Related lists and reference fields provide navigation without duplicating the same request information across every record.
- This is more maintainable than creating a single large form containing all vendor and quotation fields.
- The fourth challenge is notification quality.
- Too many notifications can create noise, while too few can cause delays. The design therefore focuses notifications on events that require action or communicate an important status change.
- Message content should identify the request and next action clearly.
- The final design challenge is preserving human responsibility. Automation should not create the impression that the system has independently evaluated suppliers. The interface and documentation explicitly identify quotation review, negotiation, and final vendor selection as procurement activities.
- The workflow automates the surrounding administrative steps.

PERFORMANCE, SECURITY AND MAINTAINABILITY

- Performance is an important quality requirement because procurement users should be able to open forms, search records, review quotations, and update workflow information without unnecessary delay.
- The implementation should keep forms focused on the information required for the current activity and avoid unnecessary client-side processing.
- Lists should display useful fields and use appropriate filters so that procurement users can locate open requests and pending tasks efficiently.
- Security is implemented through roles, ACLs, controlled record access, and workflow permissions. HR users should be able to create and track their requirements without automatically receiving access to sensitive vendor negotiations.
- Procurement users require access to commercial information, while approvers require access to the information necessary for authorization. Administrative configuration should remain restricted to ServiceNow administrators. This separation supports confidentiality and accountability.
- Maintainability is improved through meaningful table and field names, reusable Flow Designer actions, clear status values, and documented configuration. The workflow should be understandable to another administrator who did not originally build it.
- Client Scripts should be used only where configuration features are insufficient, and server-side rules should contain clearly documented business logic. Reusable subflows can reduce duplicated workflow configuration when the solution is extended to other IT procurement categories.
- Change management is also part of maintainability. Before modifying a flow, ACL, Client Script, or UI Policy, the administrator should identify the affected business stages and test cases. After the change, the complete procurement lifecycle should be retested.
- This practice reduces regression risk and provides a repeatable operational method for maintaining the system after deployment.
- The combined performance, security, and maintainability approach ensures that the application is not only functional during demonstration but also structured as a professional workflow solution.
- A reliable procurement system must protect commercial information, provide acceptable response times, and remain understandable and supportable as the organization adds more requests, vendors, and procurement categories.

FUTURE IMPROVEMENTS

- The current project establishes the core procurement workflow, but several enhancements can be added. A supplier portal could allow approved vendors to receive RFQs and submit quotation information through a controlled interface.
- This would reduce manual data entry and create a direct supplier interaction layer. Vendor users would need carefully restricted access so that each supplier could view only its own RFQs and submissions.
- Document management is another useful enhancement. Quotation documents, approvals, purchase orders, and supporting business documents could be stored with appropriate attachments or integrated document repositories.
- Version control would be useful when vendors submit revised quotations. This would make the commercial record more complete while maintaining the structured fields required for reporting.
- Analytics and dashboards could provide procurement management with indicators such as request volume, average approval time, quotation response rate, pending negotiation tasks, and purchase order cycle time.
- These measures could help identify process bottlenecks. The project should use clearly defined metrics so that reports represent the actual workflow states rather than subjective assessments.
- Integration with enterprise purchasing, finance, asset management, or inventory systems could extend the lifecycle beyond the purchase order.
- Once laptops are received, the purchase order could be linked to asset records and assignment processes. This would connect procurement with IT asset management and provide end-to-end visibility from business requirement to asset allocation.
- Advanced automation can also be introduced carefully. Rules can route approvals based on configurable policy values, remind users about overdue tasks, and detect incomplete procurement records.
- Any future analytics or recommendation capability should support procurement judgment rather than silently making commercial decisions.

ADVANCED WORKFLOW AND REPORTING ENHANCEMENTS

- Future versions can introduce reusable subflows so that common procurement actions are configured once and used by multiple procurement categories.
- **For example**, a standard approval subflow could receive a request and approval group as inputs and return an approval result. A notification subflow could standardize message formatting and recipient selection. This reduces duplicated configuration and makes future maintenance easier.
- Dynamic approval routing is another improvement. The system can use configurable organizational rules to determine whether one or more approvals are required.
- The routing logic can consider factors such as purchase value, department, or procurement category, provided those rules are defined by organizational policy. The application should store the approval chain so that the authorization history remains traceable.
- Reporting can be expanded from basic status lists to management dashboards.
- Useful views include open requests by department, requests waiting for procurement action, quotation response status, pending approvals, and purchase orders by period. Cycle-time reporting can show how long requests spend in each stage.
- These reports can help management identify process delays and evaluate whether workflow automation is reducing administrative effort.
- A further enhancement is exception monitoring. The system can identify requests that have remained in a state longer than an approved business threshold and create a reminder task or notification.
- This is a practical use of automation because it supports human follow-up rather than replacing the underlying decision-making process.

End-to-End Procurement Lifecycle

Figure 7.1 Final procurement lifecycle represented by the completed system